esade
RAMON LLULL UNIVERSITY

RECOMMENDER SYSTEMS

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Distance Metrics
for Collaborative Filtering

1. Pearson Correlation Coefficient
2. Cosine Correlation Coefficient
3. Jaccard Similarity Coefficient
4. Constrained Pearson Correlation
5. Euclidean distance Measure
6. Manhattan distance

"A Survey of Similarity Measures for Collaborative Filtering-Based Recommender System" by Gourav Jain, Tripti Mahara, Kuldeep Narayan Tripathi

Nomenclature…

Notation	Description
$r_{u_{a,i}}$	The rating given by user u_a on item i
$r_{u_{b,i}}$	The rating given by user u_b on item i
$r_{\overline{u_a}}$	The average rating value of user u_a
$r_{\overline{u_b}}$	The average rating value of user u_b
n	Total number of items
i'	Total number of co-rated items
r_{med}	Median value in rating scale
$\lvert I_{u_a} \rvert$	The cardinality of items rated by user u_a
$\lvert I_{u_b} \rvert$	The cardinality of items rated by user u_b

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Pearson Correlation Coefficient

$$\mathrm{Sim}(u_a, u_b) = \frac{\sum_{i=1}^{i'} \left(r_{u_{a,i}} - r_{\overline{u_a}}\right)\left(r_{u_{b,i}} - r_{\overline{u_b}}\right)}{\sqrt{\sum_{i=1}^{i'} \left(r_{u_{a,i}} - r_{\overline{u_a}}\right)^2} \cdot \sqrt{\sum_{i=1}^{i'} \left(r_{\overline{u_{b,i}}} - r_{\overline{u_b}}\right)^2}}$$

Linear correlation between users/items and is represented by the ratio of covariance of two users to the standard deviation between them (only considering co-rated items).

Low (high) Similarity regardless of similar (difference) in the ratings.

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Cosine Correlation Coefficient

$$\text{Sim}(u_a, u_b) = \frac{\sum_{i=1}^{i'} \left(r_{u_{a,i}}\right)\left(r_{u_{b,i}}\right)}{\sqrt{\sum_{i=1}^{n} \left(r_{u_{a,i}}\right)}\sqrt{\sum_{i=1}^{n} \left(r_{u_{b,i}}\right)}}$$

Cosine angles are formed between two rating vectors given by users. Higher similarity is represented by smaller value of angles and vice-versa.

High similarity despite of significance in ratings.

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Jaccard Similarity Coefficient

$$\mathrm{Sim}(u_a, u_b) = \frac{|I_{u_a}| \cap |I_{u_b}|}{\left|I_{u_a}\right| \cup \left|I_{u_b}\right|}$$

Only considers the number of common ratings between two users.

Does not consider the absolute ratings into account (sensitive to small data size with missing observations)**.**

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Constrained Pearson Correlation Coefficient (CPCC)

$$\mathrm{Sim}(u_a, u_b) = \frac{\sum_{i=1}^{i'} \left(r_{u_{a,i}} - r_{\text{med}}\right)\left(r_{u_{b,i}} - r_{\text{med}}\right)}{\sqrt{\sum_{i=1}^{i'} \left(r_{u_{a,i}} - r_{\text{med}}\right)^2} \cdot \sqrt{\sum_{i=1}^{i'} \left(r_{u_{b,i}} - r_{\text{med}}\right)^2}}$$

Sigmoid function-based Pearson Correlation Coefficient

$$\mathrm{Sim}(u_a, u_b) = \mathrm{Sim}(u_a, u_b)^{\text{PCC}} \cdot \frac{1}{1 + \exp\left(-\frac{|i'|}{2}\right)}$$

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Euclidean distance metric

$$\mathrm{Dis}(u_a, u_b) = \sqrt[p]{\sum_{i=1}^{i'} \left| r_{u_{a,i}} - r_{u_{b,i}} \right|^p}$$

Minkowski metric:
P=2 Euclidean distance,
P=1 City block distance (Manhattan distance)

$$\mathrm{Sim}(u_a, u_b) = \frac{1}{1 + \mathrm{Dis}(u_a, u_b)}$$

Distance between two n-dimensional points.

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THANK YOU!